

Joseph Penlap, PhD

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🌐 MyWebsite

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Education

- Oct 2022 – Feb 2026  **Ph.D., Automation, Signal and Image Processing**, University of Côte d'Azur, France.
- Thesis title: *Ecophysiological modelling of plant–nematode interactions: Understanding the origins and consequences of differential plant susceptibility.*
 - Advisors: Frédéric Grogard, Valentina Baldazzi, Suzanne Touzeau.
 - Affiliated with **Inria** and **INRAe** research centers.
- July 2021 – June 2022  **M.Sc. Mathematical Sciences, Data Science**, African Institute for Mathematical Sciences (**AIMS**), Cameroon.
- Thesis title: *Modelling the dynamics of a forest environment: role of water cycle.*
 - Advisor: Nathalie Verdière.
- Oct 2019 – July 2021  **M.Sc. Mathematics, Analysis**, University of Dschang, Cameroon.
- Thesis title: *Spaces of locally uniformly bounded functions and applications.*
 - Advisor: Jean Louis Woukeng
- Oct 2016 – July 2019  **B.Sc. Mathematics and computer science**, University of Dschang, Cameroon.

Employment History

- Sept 2025 – Aug 2026  **Temporary Teaching and Research Assistant (ATER)**, Polytech Nice Sophia, France.
- teaching mathematics and computer science courses,
 - supervising student projects (10+ students).
- Oct 2022 – Sept 2025  **Doctoral Researcher**, in teams: Inria (**Macbes** unit) & INRAE (ISA/**M2P2** unit), France.
- developing mechanistic mathematical models,
 - conducting fieldwork and laboratory experiments,
 - collection and analysis of crop data,
 - writing scientific reports and journal articles.
- Jan 2022 – Dec 2024  **Assistant Faculty**, Department of Mathematics, University of Côte d'Azur, France.
- acting as teaching assistant,
 - evaluating students' final projects.
- Apr 2019 – July 2021  **Research Intern**, Mathematics Research Unit (URMA), University of Dschang, Cameroon.

Publications & Preprints

1. **Penlap Tamagoua, J.**, Baldazzi, V., Grogard, F., and Touzeau, S., “Plant tolerance is explained by resource-based plant–nematode interactions,” *Mathematical Biosciences*, 2025 (In revision) ([hal-05394092](#)).
2. **Penlap Tamagoua, J.**, Touzeau, S., Grogard, F., and Baldazzi, V., “Unveiling plant tolerance mechanisms through a unified plant–nematode model,” *PLOS Computational Biology*, 2025 (Working paper).

Teaching Experience

Lecturer	 Polytech Nice Sophia , France. • Linear Algebra, 60 hours, semester2 - Winter 2025 to Spring 2026. • Supervision of projects, 30 hours, semester2 - Winter 2025 to Spring 2026. • Data Valorization, 46 hours, semester2 - Winter 2026. • Fundamentals in Analysis I & II, 60 hours, semester 1 - Fall to winter 2025.
Teaching Assistant	 University of Côte d'Azur , France. • Mathematical Analysis, 56 hours, semester2 - Fall 2023, 2024. • Linear Algebra, 32 hours, semester 1 - Spring 2023.
Instructor	 University of Dschang , Cameroon. • Introduction to Statistics, 15 hours, semester 1 - Spring 2021

Selected Talks & Posters

- 2025  **Coupling plant physiology and pest demography to understand plant-pest interactions**
[Talk] "From Data to Models", Sophia Antipolis, France.
[Talk] IPN LabMeeting, Sophia Antipolis, France.
- 2024  **Coupling plant physiology and pest demography to understand plant-pest interactions**
[Talk] 13th European Conference on Mathematical and Theoretical Biology (ECMTB), Toledo, Spain.
- 2023  **Modeling plant-nematode interactions to understand plant tolerance**
[Talk] Plant Pathology Department | Global Food Systems Institute, University of Florida, United States.
[Talk] MacBiosCore seminar, Porquerolles Island, France.
[Talk] Mathematical Population Dynamics, Ecology and Evolution (MPDEE), Marseille, France.
[Poster] 14th Dynamical Systems Applied to Biology and Natural Sciences (DSABNS), Bilbao, Spain.
- 2022  **Some results in Agro-forestry.**
[Talk] Biocore seminar on Automatic Control and Dynamical Systems Theory in Artificial Ecosystems, Porquerolles Island, France.
-  **Epistatic interactions between sickle-cell disease and alpha-thalassemia.**
[Talk] African Institute for Mathematical Sciences (AIMS), Limbe, Cameroon.

Skills

Languages	 French (native) and English (professional).
Coding	 Python, R, $\LaTeX$, Shell, Git, GitLab CI/CD, Streamlit, Gradio.
Data & Modelling	 Data analysis, Optimisation, Sensitivity analysis.
Operating system	 Linux, Cloud computing, High-performance computing (HPC).
Qualities	 Teamwork, problem solver, leadership.

Miscellaneous Experience

Awards and Grants

- 2022 – 2025  **INRAE-Inria PhD Fellowship**, Inria centre at University of Côte d'Azur.
- 2023  **INRAE-Inria Travel Grant**, To attend local and international conferences.

Miscellaneous Experience (continued)

- 2021 – 2022  **MasterCard Foundation Scholarship for studies at AIMS**, MSc Scholarship at the African Institute for Mathematical Sciences (**AIMS**).
- 2018 – 2021  **President Paul Biya's academic excellence prize**, Yearly granted by the President of the republic of Cameroon to the best students.
- 2020  **National winner prize OUC Africa**, CAMES.

Certification

- 2023  **TOEFL**, Awarded by British Council IELTS.
- 2022  **Test de Connaissance du Français (TCF)**. Awarded by France Éducation International.

Volunteering

- 2023 – present  **Scientific mediation addressed to high school students**, Inria centre at University of Côte d'Azur, France.
How to use mathematics for solving real problems in Agriculture?
- 2022 – present  **Member of Digital Systems for Humans (DS4H) as a doctoral student**, Inria centre at University of Côte d'Azur, France.
- 2021 – 2022  **Designing motivational scientific quotes as part of giveback activities to society**, AIMS-Cameroon, Cameroon.